

California State University, East Bay

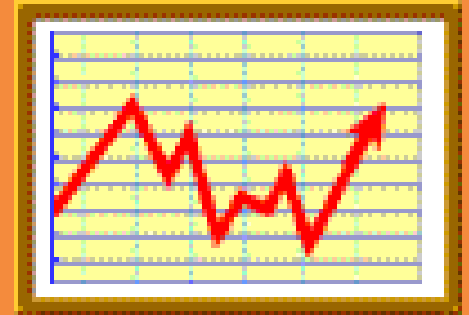
College of Business and Economics

BAN 673 Time Series Analytics

Introduction to Time Series Analytics and Forecasting Process



Lecture Materials



Dr. Z. Radovilsky

Lecture Objectives

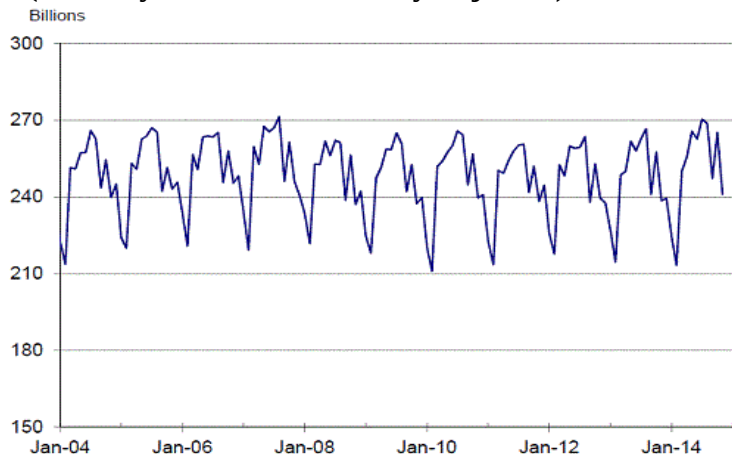
- Provide examples of time series forecasting in business
- Identify science and art of forecasting
- Recognize categories and principles of forecasting
- Present and explain steps of forecasting process
- Use of computer and software in time series forecasting



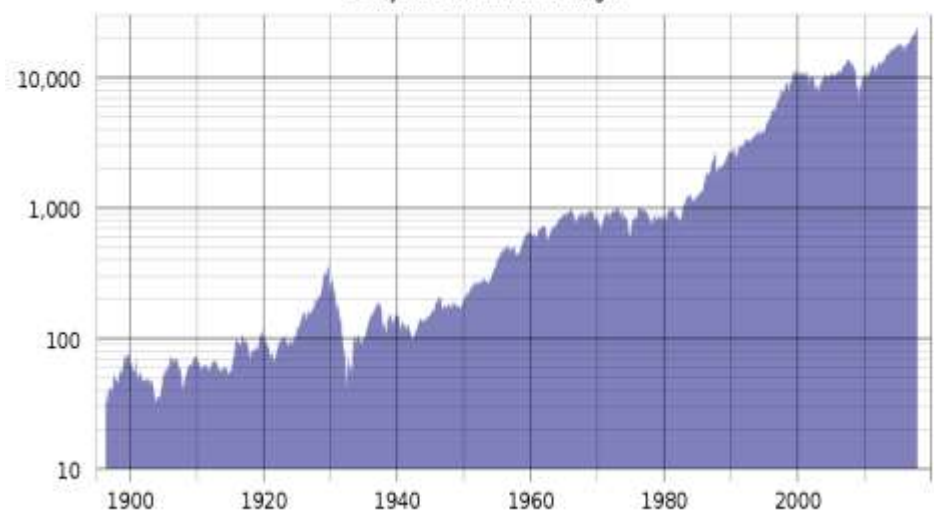
Time Series Analytics

- Time series analytics is an important part of predictive analytics dealing with making predictions by applying time series forecasting
- **Time series** is a sequence of data points taken in timely order
 - Uninterrupted set of successive data observations that have been ordered in time in equally spaced intervals - units of time, e.g., day, month, quarter, etc.
- Contrary to time series, **cross-sectional data** records are collected at the same point of time or without regard to difference in time

U.S. Highway Vehicle-Miles Traveled
(Monthly data, not seasonally adjusted)



Dow Jones Industrial Average

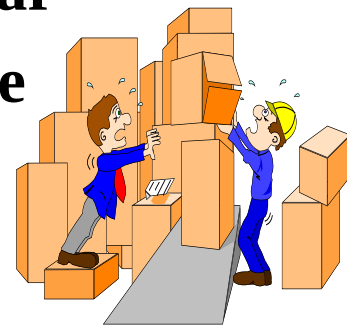


Why Time Series Forecasting in Business?

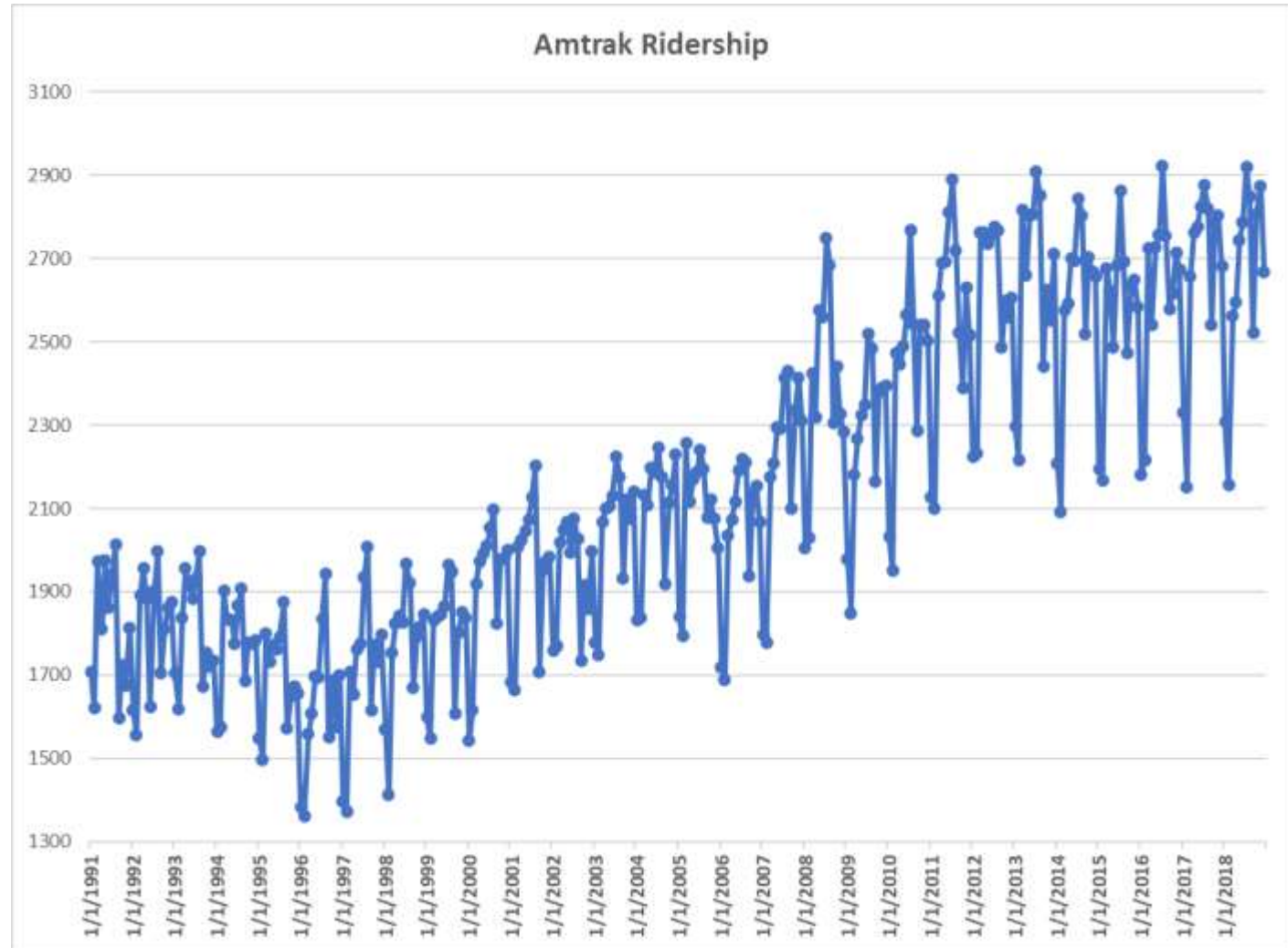
- ***Time series forecast*** means predicting the future value of a time series data
- Nothing happens until somebody forecasts something
- Forecasting is a necessary tool to reduce the uncertainty (risk) in decision making
- Better decisions come from better forecasts
- Forecasting serves as a starting point of major decisions in finance, marketing, operations, purchasing, and other functions

Decisions Requiring Time Series Forecasting

- Predicting demands of new and existing products in the following time period
- Projecting revenues, profits, expenses, and other financial results into the future
- Estimating the rate of quality improvement over a period of time
- Predicting stock prices in the next day, month or year
- Developing labor requirements over a period of time
- Predicting inventory requirements in the next year
- Forecasting number of air passengers in the future
- Forecasting ridership in U.S. railway in the future



Example: Ridership on Amtrak Trains



Basic Notation

$t=1,2,3\dots$ = time period index

y_t (Y_t) = value of the series at time period t

F_{t+k} = forecast for time period $t+k$,
given data until time t
if $k = 1$, it will be the
forecast for next period ($t + 1$) after t , F_{t+1}

e_t = forecast error (residual) for period t ,

$$e_t = y_t - F_t$$

Period t	Month	Ridership
1	1/1/1991	1708.917
2	2/1/1991	1620.586
3	3/1/1991	1972.715
4	4/1/1991	1811.665
5	5/1/1991	1974.964
6	6/1/1991	1862.356
7	7/1/1991	1939.860
8	8/1/1991	2013.264
9	9/1/1991	1595.657
10	10/1/1991	1724.924
11	11/1/1991	1675.667
12	12/1/1991	1813.863
13	1/1/1992	1614.827
14	2/1/1992	1557.088
15	3/1/1992	1891.223

Period t	Month	Ridership
325	1/1/2018	2307.547
326	2/1/2018	2157.913
327	3/1/2018	2561.693
328	4/1/2018	2595.061
329	5/1/2018	2744.626
330	6/1/2018	2787.363
331	7/1/2018	2920.412
332	8/1/2018	2848.935
333	9/1/2018	2522.367
334	10/1/2018	2807.200
335	11/1/2018	2873.968
336	12/1/2018	2668.049

Forecasting Equals Science Plus Art

■ Successful Forecasting = Science + Art

- **"Science"** implies that the body of the forecasting knowledge lies on the solid ground of quantitative forecasting methods and their correct utilization for various business situations
- **"Art"** represents a combination of a decision maker's experience, logic, and intuition to supplement the forecasting quantitative analysis. *Management abilities should be involved in forecasting process*

■ Both the science and art of forecasting are essential in developing accurate forecasts

■ All managers are forecasters

■ Better forecasts are even more critical today



Forecasting the Future Is Always Difficult...



"Heavier-than-air flying machines are impossible."

--President, Royal Society, 1895.

"Everything that can be invented has been invented."

--Commissioner, U.S. Office of Patents, 1899.

"Who the hell wants to hear actors talk?"

--Warner Brothers, 1927.

"Computers in the future may weigh no more than 1.5 tons."

--Popular Mechanics, forecasting the relentless march of science, 1949

"There is no reason anyone would want a computer in their home."

--President, Chairman and founder of Digital Equipment Corp., 1977

"Home mortgage rates will never go below 10 percent in this century."

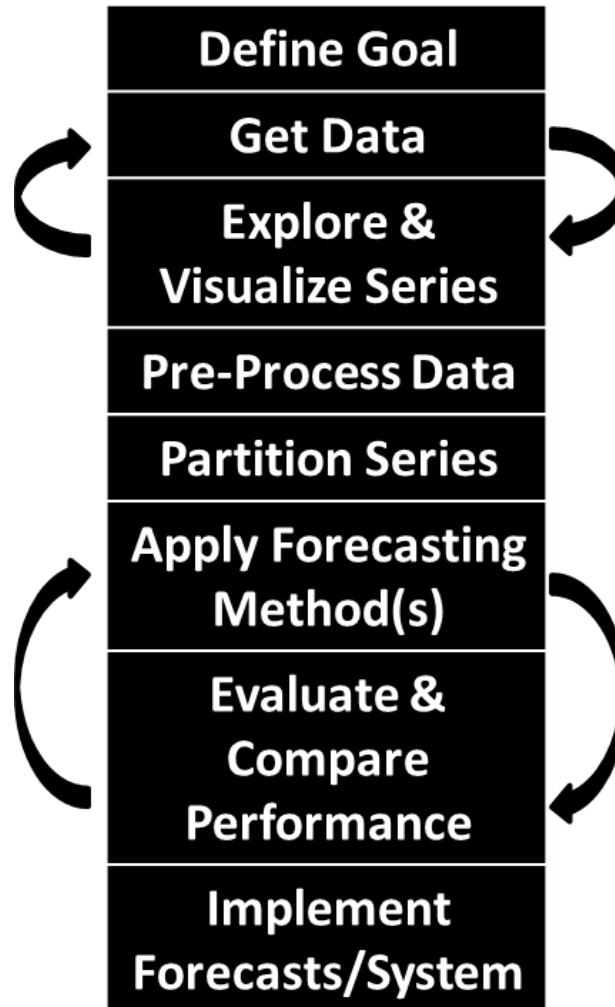
--Many financial experts, 1980

Principles of Time Series Forecasting

- Main assumption:
 - *Past pattern repeats itself into the future*
- Forecasts are rarely perfect
 - *Don't expect forecasts to be exactly equal to the actual data.*
- The science and art of forecasting try to minimize, but not to eliminate, forecast errors
 - *Forecast errors or residuals mean the difference between actual and forecasted values.*
- Forecast for a group of products (an aggregated forecast) is usually more accurate than forecasts for individual products
- Forecasts for a shorter period tend to be more accurate



Forecasting Process



Step 1: Define Goal

■ #1: Descriptive or predictive goal

- **Descriptive = time series analysis**
 - » Mean, median, mode, standard deviation, etc.
 - » Correlation of successive data points - *autocorrelation*
 - » Time series data patterns: trend, seasonality, etc.
 - » Time series visualization (data plots)
 - » Relationships to external factors
- **Predictive = time series forecasting**
 - » Predict/forecast the future time series points for which the actual time series data is not available

■ #2: Forecast horizon

- How far into the future? (k in F_{t+k})
- Long-term vs. short-term forecasting?
- Rolling forward or at single time point?
- How frequently the model should be examined?

Step 1: Define Goal

■ **#3: Forecast usage**

- Who are stakeholders and how many?
- Should forecast be numerical or binary (event/non-event)?
- Will forecast undergo adjustments?

■ **#4: Forecast expertise and automation**

- In house forecasting or consultants?
- How many series and how often will be forecasted?
- One-time forecasting or ongoing task?
- Which data and software will be available for forecasting?
- Implications:
 - » How much data is needed
 - » Choice of forecasting methods
 - » Forecasting software used
 - » Expected levels of accuracy
 - » Model deployment

Forecasting Software

- *Computer and software are critical parts of modern time series forecasting*
- **Excel** has capabilities to perform some time-series forecasting methods
 - Drawbacks of utilizing Excel
 - » No automation of the model construction process
 - » Hard to apply for complicated models
 - » Time-consuming to develop models
- General statistical (end-user) software with forecasting modules
 - **JMP, SPSS, Minitab, Stata**
- Specialized (end-user) forecasting packages for various business functions, e.g.
 - **Forecast Pro** – for sales and demand forecasting, operations and inventory management
 - **Amazon Forecast** – part of AWS platform
- General statistical and programming languages
 - **R, Python, C++**
- Time series analytics with R
 - <https://cran.r-project.org/web/views/TimeSeries.html>

